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Technical proposal (Envelope 1)

Tender NGPA/T/2026/052 · Air Quality and Noise Monitoring Network

To: The Chair, Tender Committee, Northern Gulf Ports Authority

Having examined the tender documents for NGPA/T/2026/052, we, Sabkha Environmental Systems W.L.L., submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Khalid Al-Mansoori
Managing Director, Sabkha Environmental Systems W.L.L.

Date: 8 December 2026
Reference: SES/NGPA/052/26

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1. Company and references

Sabkha Environmental Systems was founded in 2006 in Al-Noor and has 64 staff. We design, install and operate air-quality and noise monitoring networks for industrial cities, municipalities and ports. We are the authorised partner in the State for the Vireo Instruments analyser range, and we run a calibration laboratory in Al-Noor accredited to ISO/IEC 17025 (accreditation no. NGAC-CAL-0417).

Reference	Scope	Year
Ras Hadd Industrial City, ambient air network	8 stations, data platform, 5-year operation and maintenance	2021
Al-Waha Municipality, air quality and noise	5 stations, 6 noise terminals, public dashboard	2023
Eastern Province Ports, boundary monitoring	4 stations, 3 noise terminals, permit reporting	2025

2. Technical solution

Air-quality stations. Each of the six stations contains:

- a Vireo PM-2 dual-channel beta-attenuation monitor for PM10 and PM2.5, certified by TÜV as equivalent to the EN reference method (EN 16450); certificates in Appendix B;
- a Vireo NX-1 chemiluminescence NO2 analyser (EN 14211), a Vireo SX-1 UV-fluorescence SO2 analyser (EN 14212) and a Vireo OX-1 UV-photometric O3 analyser (EN 14625), all type-approved;
- a 10 m meteorological mast with an ultrasonic wind sensor, temperature, humidity and pressure sensors;
- a data logger that stores 30 days of minute data and resends it automatically after an outage, with a 4G router using two SIM cards from different operators.

Shelters. Insulated GRP shelters with two redundant air-conditioning units, rated IP55, for ambient temperatures from 0 °C to +50 °C. A UPS keeps each station running for 1 hour after a power cut and shuts the analysers down safely.

Noise terminals. SonoGuard N1 Class 1 sound level meters to IEC 61672-1, with an all-weather microphone kit (wind screen, bird spike, rain protection). Each terminal stores 1-second LAeq values and 1/3-octave spectra and sends them every minute.

Data capture. We guarantee at least 90 % valid hourly data for every pollutant in every calendar quarter. Our Al-Waha network averaged 96 % in 2025.

EnviroDesk platform. Automatic validation (range, rate of change, flat-line) and manual validation with a full audit trail. Limit-value alarms go to the HSE duty officer by SMS and e-mail within 2 minutes of the end of the averaging period. Monthly and annual reports follow the permit format and export to Excel and PDF. The public dashboard is in Arabic and English, and the published data is available through a REST API documented in OpenAPI.

3. Security and data

Hosting. EnviroDesk is installed on two virtual servers in the Authority's data centre, with a replica in its disaster-recovery site. No data leaves the Authority's infrastructure.

- Users sign in with their Active Directory accounts.

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- Raw and validated data are kept for 10 years; calibration records for 5 years.
- The stations connect to the data centre over a private APN with encrypted tunnels.

4. Implementation and timeline

Milestone	Month
Site surveys, permits and foundations	1–2
Shelters and analysers delivered to site	3
Stations and noise terminals commissioned	4–5
30-day trial run	5–6
Acceptance	6

Training. 5 days for the Authority's 6 HSE staff, in Arabic and English, covering station operation, data validation and reporting. Manuals in both languages.

5. Support, warranty and SLA

Warranty. 24 months from acceptance on all equipment and software.

Operation and maintenance for 3 years. Monthly preventive visits, calibration of every analyser every 3 months with certified gases and transfer standards traceable to our accredited laboratory, yearly verification of the noise terminals, consumables and repairs. Calibration records are kept for 5 years.

Response. An engineer is on site within 24 hours of a station failure. One spare analyser of each type is kept in our Al-Noor store.

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Annex A. Compliance matrix (Annex 1 of the RFP)

Clause	Requirement	Response	Offered
2.1	Six AQ stations: PM10, PM2.5, NO2, SO2, O3 + meteorology	Comply	§2
2.2	Four noise terminals	Comply	§2
3.1	PM analysers certified equivalent (EN 16450)	Comply	TÜV EN 16450 certificates, Appendix B (§2)
3.2	Gas analysers to EN reference methods	Comply	§2
3.3	Noise Class 1, 1/3-octave, audio recording of exceedances	Comply	SonoGuard N1 Class 1 (§2)
3.4	Shelter IP55, 0 to +50 °C, UPS ≥ 2 h	Comply	IP55, 0 to +50 °C, UPS 1 h (§2)
3.5	7 days local storage, back-fill, backup link	Comply	30 days, dual-SIM 4G (§2)
3.6	≥ 90 % valid hourly data per quarter	Comply	90 % guaranteed (§2)
4.1	Authority data centre; 10-year retention	Comply	Authority data centre + DR site (§3)
4.2	Validation with audit trail	Comply	§2
4.3	Alarms within 5 min (SMS, e-mail)	Comply	Within 2 minutes (§2)
4.4	Monthly / annual permit reports (Excel, PDF)	Comply	§2
4.5	Public dashboard AR/EN + REST API	Comply	§2
4.6	Active Directory sign-in	Comply	§3
5.1	In operation ≤ 6 months; 30-day trial	Comply	Acceptance month 6 (§4)
5.2	Calibration every 3 months, ISO/IEC 17025 traceable	Comply	Every 3 months, accredited lab (§5)
5.3	48 h on-site response; spares in the State	Comply	24 h, spares in Al-Noor (§5)
5.4	24-month warranty; 3-year O&M	Comply	§5
5.5	Training of 6 HSE staff, AR/EN	Comply	§4
6.1	≥ 3 similar networks in 7 years	Comply	3 networks (§1)
6.2	Office or service partner in the State	Comply	Office in Al-Noor (§1)